

- » Making a network plan
- » Taking stock of your computer stock
- » Making sure that you know why you need a network
- » Making the basic network decisions that you can't avoid
- » Looking at additional topics your plan should address

Chapter 1

Local Area Networks

Okay, so you're convinced that you need to network your computers. What now? Do you stop by Computers-R-Us on the way to work, install the network before drinking your morning coffee, and expect the network to be fully operational by noon?

I don't think so.

Networking your computers is just like any other worthwhile endeavor: Doing it right requires a bit of planning. This chapter helps you to think through your network before you start spending money. It shows you how to come up with a networking plan that's every bit as good as the plan that a network consultant would charge thousands of dollars for. See? This book is already saving you money!

Making a Network Plan

Before you begin any networking project, whether a new network installation or an upgrade of an existing network, make a detailed plan *first*. If you make technical decisions too quickly, before studying all the issues that affect the project, you'll regret it. You'll discover too late that a key application won't run over the network, the network has unacceptably slow performance, or key components of the network don't work together.

Here are some general thoughts to keep in mind while you create your network plan:

- » **Don't rush the plan.** The most costly networking mistakes are the ones that you make before you install the network. Think things through and consider alternatives.
- » **Write down the network plan.** The plan doesn't have to be a fancy, 500-page document. If you want to make it look good, pick up a ½-inch three-ring binder, which is big enough to hold your network plan with plenty of room to spare.
- » **Ask someone else to read your network plan before you buy anything.** Preferably, ask someone who knows more about computers than you do.
- » **Keep the plan up to date.** If you add to the network, dig up the plan, dust it off, and update it.



TIP

“The best laid schemes of mice and men gang aft a-gley, and leave us naught but grief and pain for promised joy.” Robert Burns lived a few hundred years before computer networks, but his famous words ring true. A network plan isn't chiseled in stone. If you discover that something doesn't work the way you thought it would, that's okay. Just change your plan.

Being Purposeful

One of the first steps in planning your network is making sure that you understand why you want the network in the first place. Here are some of the more common reasons for needing a network, all of them quite valid:

- » **My co-worker and I exchange files using flash drives just about every day.** With a network, trading files is easier.
- » **I don't want to buy everyone a color laser printer when I know the one we have now just sits there taking up space most of the day.** So wouldn't buying a network be better than buying a color laser printer for every computer?
- » **I want everyone to be able to access the internet.** Many networks, especially smaller ones, exist solely for the purpose of sharing an internet connection.
- » **Business is so good that one person typing in orders eight hours each day can't keep up.** With a network, more than one person can enter orders, which expedites orders and possibly saves on overtime expenses.

- » **My brother-in-law just put in a network at his office.** No one wants to be behind the times.
- » **I already have a network, but it's so old that it may as well be made of kite string and tin cans.** An improved network speeds up access to shared files, provides better security, is easier to manage, and is more reliable.



REMEMBER

After you identify all the reasons why you think you need a network, write them down. Don't worry about winning the Pulitzer Prize for your stunning prose. Just make sure that you write down what you expect a network to do for you. If you were making a 500-page networking proposal, you'd place the description of why a network is needed in a tabbed section labeled Justification. In your 1/2-inch network binder, file the description under Purpose.



TIP

As you consider the reasons why you need a network, you may conclude that you don't need a network after all. That's okay. You can always use the binder for your stamp collection.

Taking Stock

One of the initial challenges of planning a network is figuring out how to work with the computers that you already have. In other words, how do you get from here to there? Before you can plan how to get "there," you have to know where "here" is. In other words, you have to take a thorough inventory of your current computers.

What you need to know

You need to know the following information about each of your computers:

- » **The processor type and, if possible, its clock speed:** It would be nice if each of your computers had a shiny new i7 12-Core processor. In most cases, though, you find a mixture of computers: some new, some old, some borrowed, some blue.



TIP

You can't usually tell what kind of processor a computer has just by looking at the computer's case. But you can easily find out by right-clicking Computer on the Start menu and choosing Properties.

» **The size of the hard drive and the arrangement of its partitions:** To find the size of your computer's hard drive, open the Computer window, right-click the drive icon, and choose the Properties command from the shortcut menu that appears. Figure 1-1 shows the Properties dialog box for a 476GB hard drive that has about 224GB of free space.

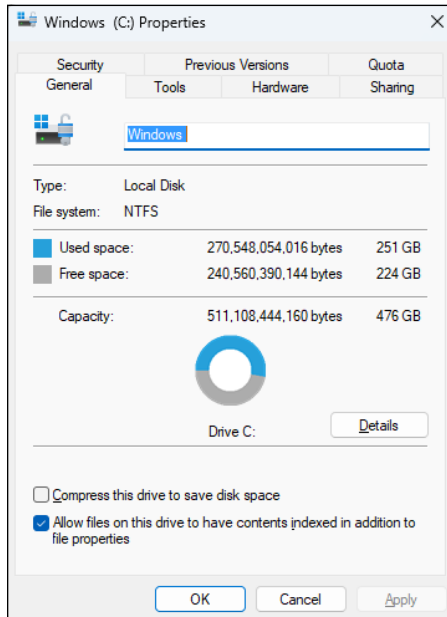
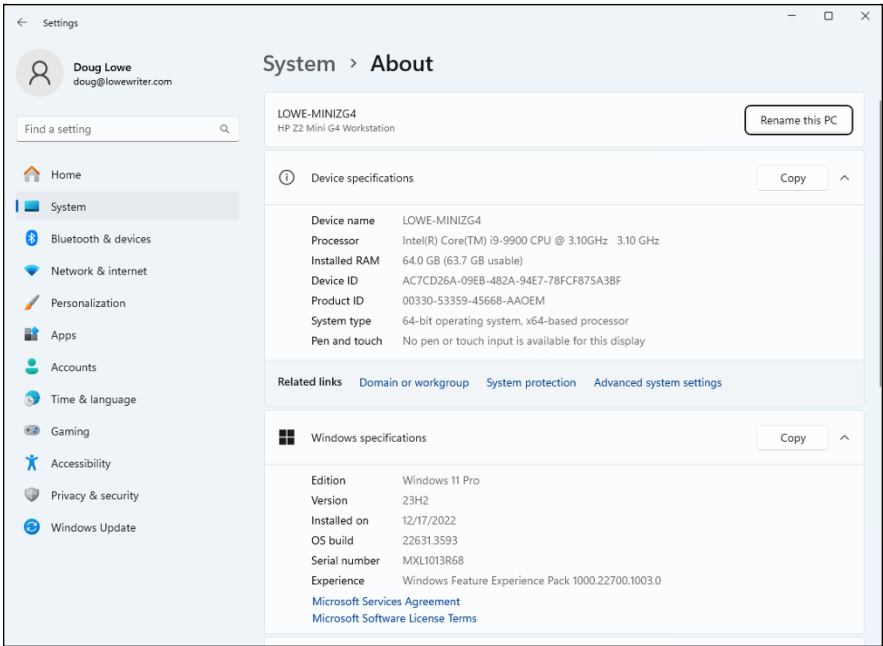


FIGURE 1-1:
The Properties
dialog box for a
disk drive.

If your computer has more than one hard drive, Windows lists an icon for each drive in the Computer window. Jot down the size and amount of free space available on each drive.

- » **The amount of memory:** To find this information in Windows, right-click Computer from the Start menu and choose the Properties command. The amount of memory on your computer is shown in the dialog box that appears. For example, Figure 1-2 shows the System Properties page for a computer with 64GB of RAM.
- » **The operating system version:** This you can also deduce from the System Properties dialog box. For example, the Properties page shown in Figure 1-2 indicates that the computer is running Windows 11 Professional.

FIGURE 1-2:
The Properties
page for a
computer with
64GB of RAM.



- » **What kind of printer, if any, is attached to the computer:** Usually, you can tell just by looking at the printer. You can also tell by double-clicking the Printers icon in Control Panel.
- » **Any other devices connected to the computer:** A DVD or Blu-ray drive? Scanner? External disk? Webcam? Battle droid? Hot tub?
- » **What software is used on the computer:** Microsoft Office? AutoCAD? QuickBooks? Make a complete list and include version numbers.

Programs that gather information for you

Gathering information about your computers is a lot of work if you have more than a few computers to network. Fortunately, several software programs are available that can automatically gather the information for you. These programs inspect various aspects of a computer, such as the CPU type and speed, amount of RAM, and the size of the computer's hard drives. Then they show the information on the screen and give you the option of saving the information to a hard drive file or printing it.

Windows comes with just such a program: Microsoft System Information. Click Start, type **System Information**, and click the link to the System Information program.

When you fire up Microsoft System Information, you see a window similar to the one shown in Figure 1-3. Initially, Microsoft System Information displays basic information about your computer, such as your version of Microsoft Windows, the processor type, the amount of memory on the computer, and so on. You can obtain more detailed information by clicking Hardware Resources, Components, or other categories in the left side of the window.

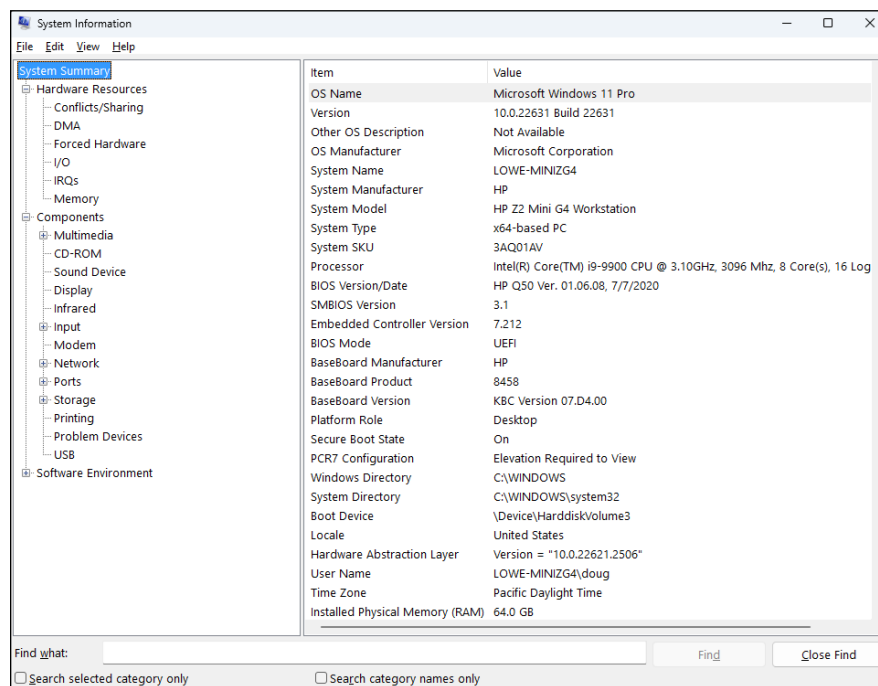


FIGURE 1-3:
Let the System
Information
program gather
the data you
need.

Considering Cable

Over the years, several different types of cables have been used for networking. But today, almost all cabled networks are built using simple copper-based unshielded twisted pair (UTP) cable. Figure 1-4 shows a twisted pair cable.

When you use UTP cable to construct an Ethernet network, you connect the computers in a starlike arrangement, in which each computer is connected to a central point. At the center of the stars are switches (see Book 1, Chapter 3). Depending on the model, a single switch can connect from 4 to 48 or more devices.

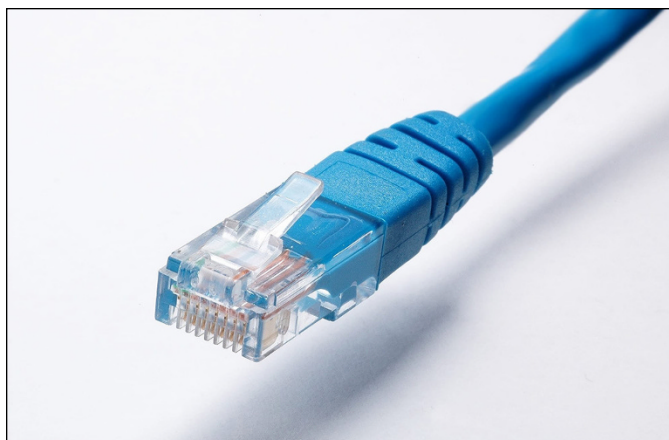


FIGURE 1-4:
Twisted-pair
cable.

Here are a few additional details that you should know about twisted pair cabling:

- » UTP cable consists of four pairs of thin wire twisted around each other; several such pairs are gathered up inside an outer insulating jacket. Ethernet uses two pairs of wires, or four wires altogether.
- » UTP cable comes in various grades known as *categories*. Don't use anything less than Category 5e cable for your network; Category 6 is better yet. Although lower-category cables may be less expensive, they won't be able to support faster networks.



TIP

Be prepared for the future. Although higher-category cables are more expensive than lower-category cables, the real cost of installing Ethernet cabling is the labor required to actually pull the cables through the walls. As a result, I recommend that you invest in Category 6.

- » UTP cable connectors look like modular phone connectors but are a bit larger. UTP connectors are officially called RJ-45 connectors.
- » UTP cable can be purchased in prefabricated lengths, but for most jobs you'll want to purchase the cable in bulk and have a professional installer attach the connectors. Or, you can attach the connectors yourself using a simple crimping tool you can purchase for about \$50.
- » The maximum allowable cable length between the switch and the computer is 100 meters (about 328 feet). That should be more than enough for most circumstances, but don't forget that the distance includes the vertical distance required for getting from the floor up to the ceiling, and back down again.
- » Always leave at least 5 feet or more of extra cable neatly coiled up in the ceiling space above each location where the cable drops through the wall to floor level. That way, you'll have some flexibility to re-route the cable later on if necessary.

Surmising Switches

As I mention in the previous section, computers and other devices are connected to a network in a starlike configuration, with switches at the center of the star. Figure 1-5 shows a switch with five computers connected to it.

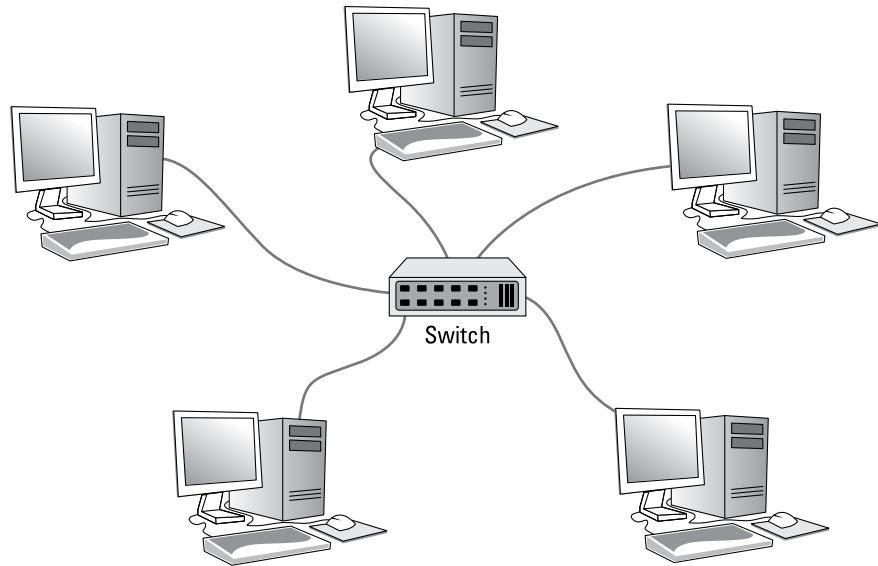


FIGURE 1-5:
A switch with
five computers
connected.

A switch contains a number of *ports*, each of which is a receptacle that can accommodate an RJ-45 jack connected to a UTP cable. In Figure 1-5, there are five UTP cables. One end of each of these cables is plugged into a port on the switch, and the other end is plugged into the computer's network adapter.

Although it may not be obvious from the figure, the switch does not have to be in the same room as the computers. In fact, ideally the switch will be in a separate room from the computers. The cables run through the ceilings and the walls from the location of the switch to the location of the computers, within the 100-meter limit of UTP cable. (The switches are generally located in the same room as the servers.)

Here are some additional ins and outs for working with switches:

- » Because you must run a cable from each computer to the switch, find a central location for the switch to which you can easily route the cables.



WARNING

- » The switch requires electrical power, so make sure that an electrical outlet is handy.
- » As a general rule, purchase twice as many switch ports as you currently need. Don't buy an eight-port switch if you want to network eight computers because when (not *if*) you add the ninth computer, you'll have to buy another switch.
- » You can connect — or *daisy-chain* — switches to one another, as shown in Figure 1-6. You connect one end of a cable to a port on one switch and the other end to a port on the other switch.
- » Although you can daisy-chain as many switches together as you want, in actual practice you should limit the number of daisy chains in your switch configuration. Daisy-chaining can slow down a network a bit because each switch must fully receive each packet before it begins to forward the packet to the next switch. (However, some switches actually start the packet forwarding before the entire packet is received, which reduces the performance hit a bit.)
- » If you need more ports than a single switch can provide, you can use *stackable switches*. Stackable switches have high-speed direct connections that enable two or more switches to be connected in such a way that they behave as if they were a single switch.

This type of connection is sometimes called a *back-plane connection* because the interconnect may be on the back of the switch, but that's not always the case. If a single switch will suffice for you now, but there is a reasonable chance that you might outgrow it and need a second switch, I suggest you invest in a stackable switch so that you can expand your network later without daisy-chaining.

- » Another way to provide a high-speed interconnect between switches is to purchase switches that have a few high-speed SFP ports. You can then equip those ports with 10 Gb connections to route traffic between the switches. (These high-speed connections can also be used to connect switches to servers.)
- » Yet another way to create high-speed interconnects between switches is to use a feature called *link aggregation*. If your switches provide this feature, you simply run two or more cables between the switches, using two or more ports on each switch. Then, you use the switch's configuration software to bond the two ports together to create one link with double the port speed.

Professional-quality network switches have network-management features that allow you to log in to the switch, usually via a web interface, to monitor and configure the switch. Such switches are called *managed switches*. Consumer-grade switches, also called *unmanaged switches*, are less expensive primarily because they do not support this feature. If you have more than a few dozen users, you'll want to invest in managed switches.

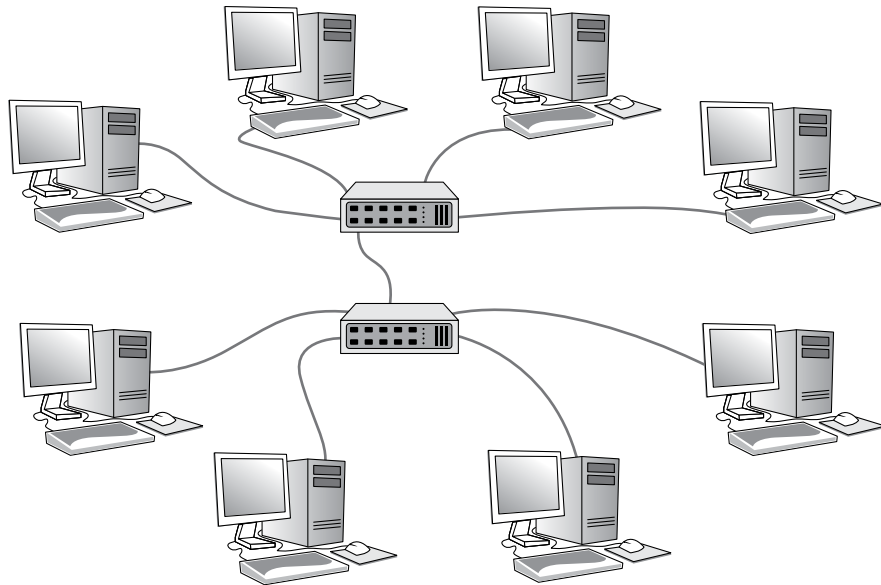


FIGURE 1-6:
Daisy-chaining
switches.

Planning the Network Topology

Topology refers to the way the devices in your network are connected to each other via network switches. You'll need to determine what kind of switches to use, how many, where to run the cable, where to locate the switches, and so on.

Here are just a few of the questions to consider:

- » Where will the servers be located? Will there be just a single switch in the server room, or will there be a central switch with workgroup switches located closer to the computers?
- » Where will you place workgroup switches? On a desktop somewhere within the group or in a small wiring closet?
- » How many client computers will you place on each workgroup switch, and how many switches will you need?
- » If you need more than one switch, how will you connect the switches to each other?
- » If you need wireless networking, what kind of wireless networking devices will you need, where will you put them, and how will you connect them to the network?

For midsized networks (say, 50 to 200 users), a common way to design the network topology is to use a two-layer switch architecture as shown in Figure 1-7:

- » **Core layer:** The *core layer* contains high-performance switches that connect to the servers, the internet gateway, and to each other. These connections should be as fast as possible — ideally, 10 Gbps fiber or copper connections using SFP ports.
- » **Access layer:** The *access layer* consists of switches that are connected to the core layer and to the end-user computers.

In Figure 1-7, there is one switch at the core layer and four switches at the access layer. The two core switches are connected to each other, to the servers, and to the access layer switches using 10 Gbps fiber SFP connections. The access switches connect to the computers using standard 1 Gb Ethernet connections.

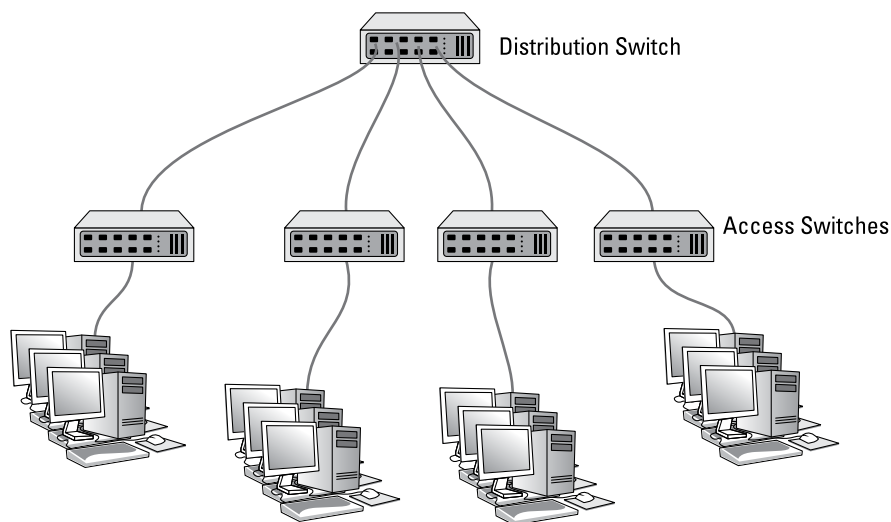


FIGURE 1-7:
A two-tiered
switch design.

For even larger networks, a three-tier design can be used. In that case, a *distribution layer* is added between the access and core layers. The servers are moved to the distribution layer and the core layer using specialized high-speed switches whose sole purpose is to move large amounts of data between the distribution switches as quickly as possible.

Planning the TCP/IP Implementation

In addition to planning for the physical parts of your network infrastructure (cables, switches, ports, and so on), you'll also need to plan the details of how you'll implement TCP/IP for your network. TCP/IP is the basic networking protocol that your network uses to keep track of the individual computers and other devices on the network. Each computer or device will need an IP address (for example, 10.0.101.65). You'll need to devise a plan for how these addresses will be allocated.

You learn everything you need to know about TCP/IP in Book 2, Chapter 2, so make sure you understand the information in that chapter before you complete this part of your plan. For now, here are some of the main points your plan should address:

» **The subnet and VLAN structure of your network:** Will everything be on a single subnet, or will you use two or more subnets to separate different types of devices?

Although it isn't impossible, dividing an existing network into separate subnets later on is a bit of a pain. So unless your network is very small, I suggest you plan on using subnets from the very start. In particular, you should consider using separate subnets for the following:

- **Wireless networks:** In fact, if you create two or more wireless networks (for example, a corporate network for company-owned mobile devices, an employee network for personal smartphones, and a guest network for visitors), I suggest a separate subnet for each of the wireless networks.
- **Remote locations that will be connected via a VPN tunnel.**

In addition, if you use IP phones, definitely put the phones on their own subnet. And if your organization has more than a few dozen users, consider dividing them among two or more subnets according to their work groups. For example, you might use one subnet for the sales department and another one for the production department.

You'll probably need to set up VLANs to manage your subnets. Typically, there's a one-to-one correspondence between subnets and VLANs; in other words, each subnet lives on its own VLAN.

Don't go overboard with the subnets, however. Try to find the right balance between running the entire organization on a single subnet versus creating a lot of subnets, each with just a few users.

Why bother with the subnets? The main reason is to avoid issues that will come up when your organization grows. You may have just 20 employees now, but years from now, when you have 100, and everyone starts bringing their smartphones and tablets and connecting to your Wi-Fi, you'll find that



TIP



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the limit of 253 devices per subnet on a 255.255.255.0 network is simply not enough. When you run out of DHCP space and your users can't get on the network, you'll wish you had spread things out over a couple of subnets.

- » **The DHCP structure:** Speaking of DHCP, what server will be responsible for DHCP? What will be the DHCP scope — that is, the range of addresses that are given out by DHCP? How will the size of your scope accommodate all the devices that will require DHCP on the subnet, with plenty of room for growth? (Again, be especially careful if you connect a wireless access point to the same subnet that your cabled network is on. You'll be surprised how quickly you can run out of IP addresses when every one of your users brings an iPhone and an iPad. Subnets are the best answer to that problem.)
- » **The static IP addresses of devices whose IP should never change:** These devices may include servers, printers, firewalls, and other managed devices. You'll be surprised how quickly these can add up, as well. You'll need static IP addresses for each of the network interfaces on your servers, for your switches, printers, copiers, fax machines, firewalls, routers, tape backup devices, and network storage devices. If you use virtualization software, the host processors will also need an IP address for each network interface. Even your UPS battery backups may want an IP address. The list goes on and on.



TIP

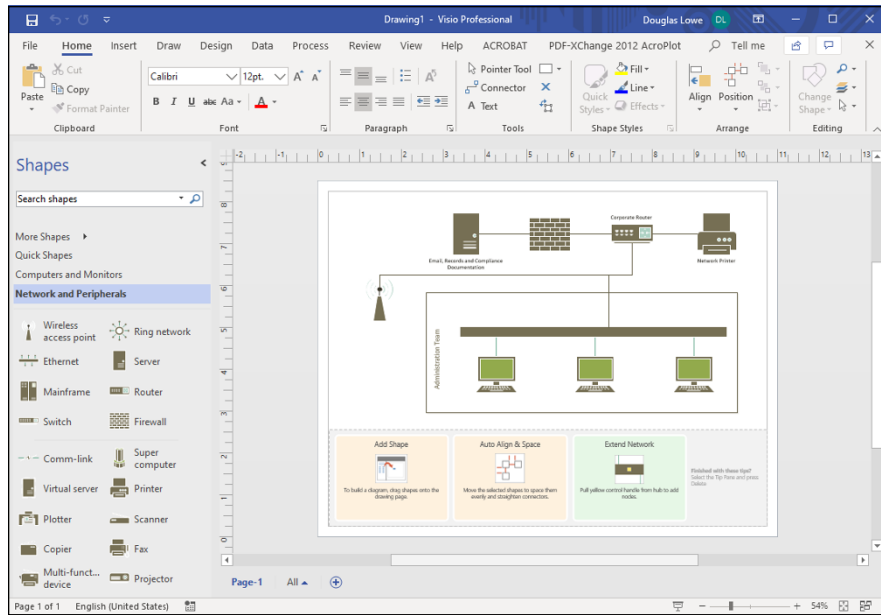
It is absolutely imperative that you keep a good record of what static IP addresses you have assigned in your network, and that you configure your DHCP server properly so that it doesn't step on top of static IP addresses. Every time you add a device with a static IP address, be sure to update your list. And, just as important, whenever you retire a device that uses a static IP address, update your master list to remove the IP address.

Drawing Diagrams

One of the most helpful techniques for creating a network plan is to draw a picture of it. The diagram can be a detailed floor plan, showing the actual location of each network component: a *physical map*. If you prefer, the diagram can be a *logical map*, which is more abstract and Picasso-like. Any time you change the network layout, update the diagram. Also include a detailed description of the change, the date that the change was made, and the reason for the change.

You can diagram very small networks on the back of a napkin, but if the network has more than a few computers, you'll want to use a drawing program to help you create the diagram. One of the best programs for this purpose is Microsoft Visio, shown in Figure 1-8.

FIGURE 1-8:
Using Visio to
draw a network
diagram.



Here's a rundown of some of the features that make Visio so useful:

- » Smart shapes and connectors maintain the connections drawn between network components, even if you rearrange the layout of the components on the page.
- » Stencils provide dozens of useful shapes for common network components — not just for client and server computers, but for routers, hubs, switches, and just about anything else you can imagine. If you're really picky about the diagrams, you can even purchase stencil sets that have accurate drawings of specific devices, such as Cisco routers or IBM mainframe computers.
- » You can add information to each computer or device in the diagram, such as the serial number or physical location. Then, you can quickly print an inventory that lists this information for each device in the diagram.
- » You can easily create large diagrams that span multiple pages.

More Questions Your Network Plan Should Address

In addition to the basic questions of why you need a network, what kind of servers you need to provide, and what kind of infrastructure your network will require, your network plan should address the following questions:

- » **Does it allow for growth?** What growth areas do you anticipate over the next few years? Does this network plan provide for such growth? For example, if you currently have 20 devices on the network, a 24-port switch may be adequate for today. But you should consider a 48-port switch instead. It will cost more now, but will simplify your expansion down the road.

Similarly, if you anticipate that each office will have just one employee, consider what you'll have to do if you run out of offices and end up putting two employees in each office. If you run a single cable to each office now, you'll have to pay to have a second cable run later. Better to spend a little more for extra cable and have the installer pull two cables to each office. (Better yet, have the installer pull three cables to each office: When you move a second employee into the office, you may also put a printer in there.)

- » **How will you secure it?** What kind of safety precautions will you take to keep unwanted visitors off your network? You'll need a strong, well-configured firewall to keep intruders from breaking in to your network via your internet connection. If you're installing wireless access points, you'll have to take precautions to secure the wireless networks. And you'll need strong password policies to prevent hackers who do manage to get on to your network from getting at any valuable data.

For more information about network security, refer to the chapters in Book 10.

- » **How will you back it up?** You'll need to include a solid plan to back up your servers and the data that resides on them. That plan will probably require additional hardware, such as separate disk storage to hold the first level of backup data, as well as a means to get the backed-up files off-site so they can survive a true disaster such as a fire or flood.

You'll also need to be certain that you provide adequate network disk storage so that all users can put all their work on the network, where it can be backed up. In lieu of that, you'll need a plan that backs up not only your servers, but also the client computers.

For more information about backing up your network, refer to Book 3, Chapter 6.

- » **How will you recover from failures?** Make sure you have a plan in place that will protect you from the commonplace maladies of daily life such as occasional power failures, as well as from the unforeseen, such as vandalism, theft, or fire. Every device on your network, no matter how insignificant, should be protected by battery backup. When possible, you should have spares of critical components.

For more information about disaster recovery, see Book 10, Chapter 4.

